Medical Insurance Cost Prediction

Supervised Learning - Assignment 1

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1 Introduction

This report applies linear regression techniques to predict medical insurance costs (charges) in the United States (US) healthcare system. We seek to construct a linear supervised learning model to identify key determinants of individual medical costs billed by health insurance for patients in the US. In doing so, we propose some solutions to address the following questions for stakeholders:

- **Patients**: Which personal factors most impact insurance charges?
- **Insurers**: How can risk be accurately quantified for premiums?
- **Policymakers**: Which public health initiatives can reduce costs?

The dataset includes 500 US patients with the following 6 features:

- **age**: Integer, primary beneficiary's age
- **sex**: Factor, gender of insurance contractor (female/male)
- **bmi**: Continuous, body mass index
- **children**: Integer, number of dependents
- **smoker**: Factor, smoking status (yes/no)
- **region**: Factor, US residential area (northeast, southeast, southwest, northwest)

The target variable is **charges** - medical costs billed. We aim to build linear models that predict charges based on the aforementioned characteristics, identifying key factors and interactions.

2 Data Import and Preparation

We use a sub-sample of 500 observations, randomly selected with seed 756`for reproducibility. The`.RData`file containing both the full insurance data set (`fulldata`) and sampled data (`my_data‘) can be found here.
The original dataset can be found on Kaggle, here.
Please see the GitHub repo here for the full codebase.

3 Exploratory Data Analysis

3.1 Data Overview

A sample of the dataset is shown below:

Table 1: Insurance Dataset Sample (First 5 Rows)

age	sex	bmi	children	smoker	region	charges
31	female	38.09	1	yes	northeast	58571.07
60	male	40.92	0	yes	southeast	48673.56
33	female	35.53	0	yes	northwest	55135.40
63	female	37.70	0	yes	southwest	48824.45
60	male	32.80	0	yes	southwest	52590.83

Summary statistics for numerical variables:

Table 2: Summary Statistics of Numerical Variables

Variable	Min	Q1	Median	Mean	Q3	Max
Age	18.00	27.00	39.00	39.02	51.00	64.00
BMI	16.82	26.60	30.74	30.87	34.72	48.07
Children	0.00	0.00	1.00	1.08	2.00	5.00
Charges	1121.87	4746.88	8828.19	13000.91	16818.54	58571.07

Categorical variable distributions:

Table 3: Summary of Categorical Variables

	Variable	Count	Percentage
female	Sex (female)	253	50.6
male	Sex (male)	247	49.4
no	Smoker (no)	398	79.6
yes	Smoker (yes)	102	20.4
northeast	Region (northeast)	131	26.2
northwest	Region (northwest)	116	23.2
southeast	Region (southeast)	126	25.2
southwest	Region (southwest)	127	25.4

Key characteristics:

- **Age**: 18–65 years, median 39 years
- **BMI**: Median 30.7, overweight range (25–30)
- **Children**: Mostly 0–1, max 5
- **Charges**: $1,122 to $58,571, right-skewed
- **Sex**: 50.6% female
- **Smoker**: 20.4% smokers
- **Region**: Evenly distributed

3.2 Visual Analysis

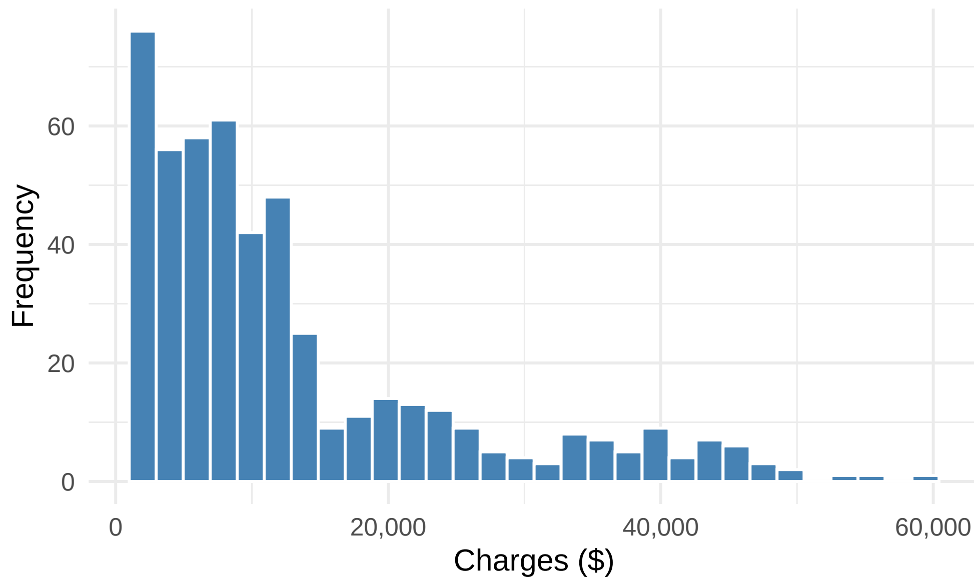

Figure 1: Distribution of Insurance Charges

Charges are right-skewed, with most values between \$5,000–\$15,000, and some exceeding \$40,000.

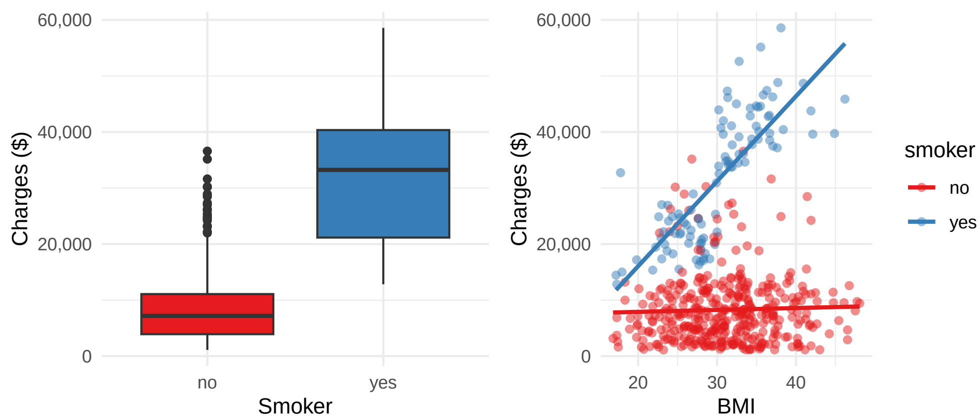

Figure 2: Charges by Smoking Status and BMI vs. Charges

Smokers face 3–4 times higher charges. BMI has a stronger effect on charges for smokers, indicating an interaction.

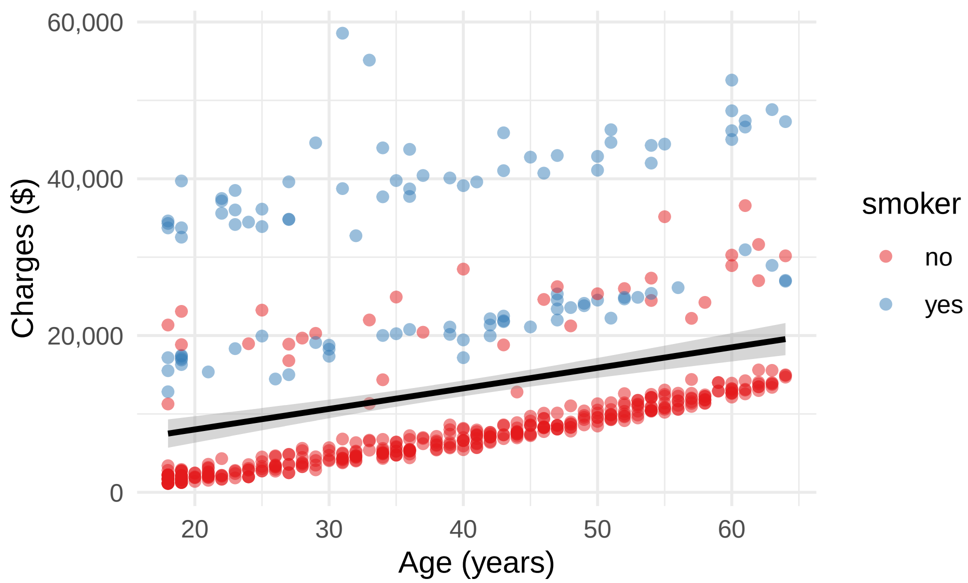

Figure 3: Age vs. Charges

Age has a positive linear relationship with charges.

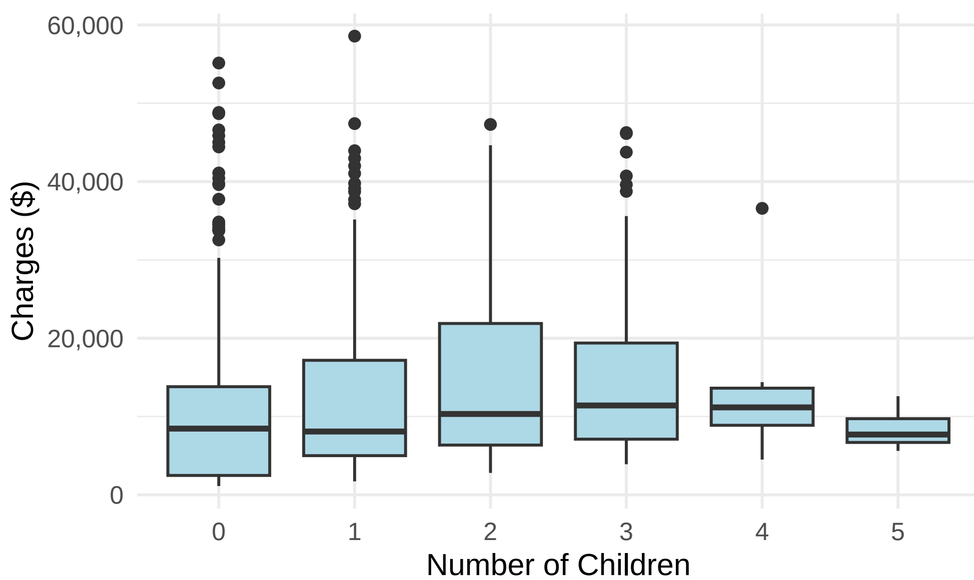

Figure 4: Children vs. Charges

More children slightly increase charges, less impactful than age or smoking.

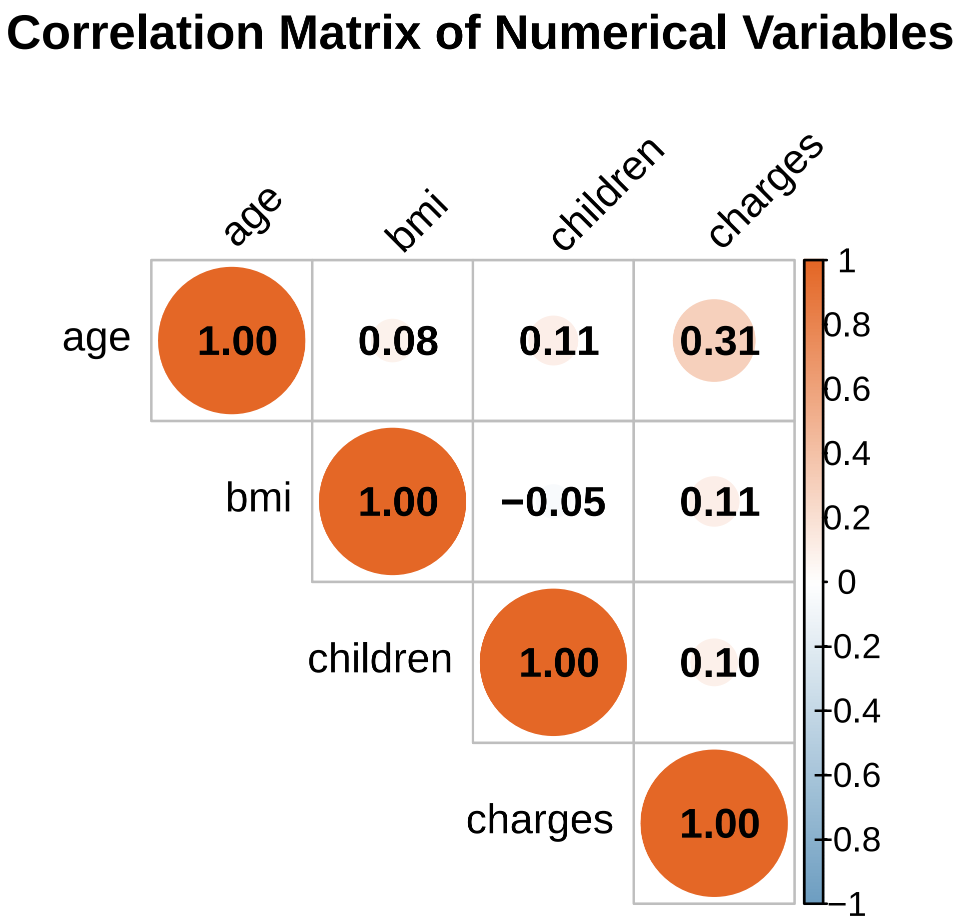

Figure 5: Correlation Matrix

Correlations: age (r = 0.31), BMI (r = 0.11), children (r = 0.1) with charges; low multicollinearity.

3.3 Key Findings

- **Smoking**: Dominant factor, significantly increasing charges
- **Age**: Linear increase in charges
- **BMI**: Stronger effect for smokers
- **Children**: Modest effect
- **Region/Gender**: Minor impact
- **Charges**: Right-skewed distribution

4 Model Building

4.1 Data Splitting

Training set:400observations (80%) of 500 in the total sample Testing set:100observations

4.2 Full Regression Model

Table 4: Full Regression Model Coefficients

	Term	Coefficient	Std. Error	t value	p value	
(Intercept)	(Intercept)	-10335.00	1912.72	**-5.40**	0.000	*
age	age	285.84	22.76	**12.56**	0.000	*
sexmale	sexmale	-243.14	620.50	-0.39	0.695	
bmi	bmi	259.02	52.85	**4.90**	0.000	*
children	children	356.43	259.40	1.37	0.170	
smokeryes	smokeryes	24127.34	769.42	**31.36**	0.000	*
regionnorthwest	regionnorthwest	-1136.93	891.64	-1.28	0.203	
regionsoutheast	regionsoutheast	-1442.71	871.88	-1.65	0.099	
regionsouthwest	regionsouthwest	-1156.14	860.13	-1.34	0.180	

$R^2 = 0.748$. Smoking, age, and BMI are significant predictors.

Test metrics: RMSE $6,129, MAE $4,243, R^2 0.684.

5 Model Improvement

5.1 Variable Selection and Regularization

We explored stepwise (AIC/BIC), Ridge, and LASSO regression.

Table 5: Feature Selection Comparison

Feature	Full Model	Stepwise (AIC)	Stepwise (BIC)	LASSO
Intercept	x	x	x	x
Age	x	x	x	x
Sex (male)	x			
BMI	x	x	x	x
Children	x			x
Smoker (yes)	x	x	x	x
Region (northwest)	x			
Region (southeast)	x			
Region (southwest)	x			

5.2 Interaction Model

We include a BMI-smoking interaction term based on our findings in Figure 2 where we uncovered that BMI's effect on charges may not be constant across smoker vs. non-smoker groups.

Table 6: Interaction Model Coefficients

	Term	Coefficient	Std. Error	t value	p value	
(Intercept)	(Intercept)	-629.00	1563.43	-0.40	0.688	
age	age	281.05	17.30	**16.24**	0.000	*
sexmale	sexmale	-598.96	472.28	-1.27	0.205	
bmi	bmi	-46.12	44.05	-1.05	0.296	
children	children	553.58	197.59	**2.80**	0.005	*
smokeryes	smokeryes	-28318.09	3154.44	**-8.98**	0.000	*
regionnorthwest	regionnorthwest	-981.69	678.04	-1.45	0.148	
regionsoutheast	regionsoutheast	-1488.89	662.96	**-2.25**	0.025	*
regionsouthwest	regionsouthwest	-1675.12	654.74	**-2.56**	0.011	*
bmi:smokeryes	bmi:smokeryes	1728.56	102.16	**16.92**	0.000	*

The interaction is significant ($p < 0.001$). BMI effect: $-46 (non-smokers), $1682 (smokers). The different slopes shown below, illustrate the interaction - BMI's effect on charges is steeper for smokers:

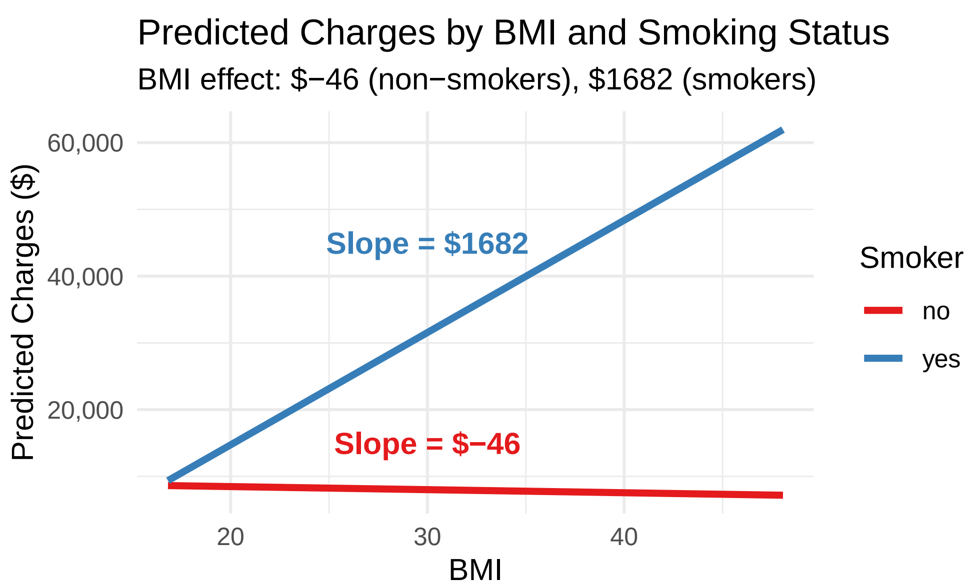

Figure 6: BMI-Smoking Interaction Effect

5.3 Model Comparison

Table 7: Model Performance Comparison

Model	RMSE	R_Squared
Interaction Model	**5689.43**	**0.728**
LASSO	5983.06	0.699
Ridge	5986.03	0.699
Stepwise (AIC)	6043.31	0.693
Stepwise (BIC)	6043.31	0.693
Full Model	6128.86	0.684

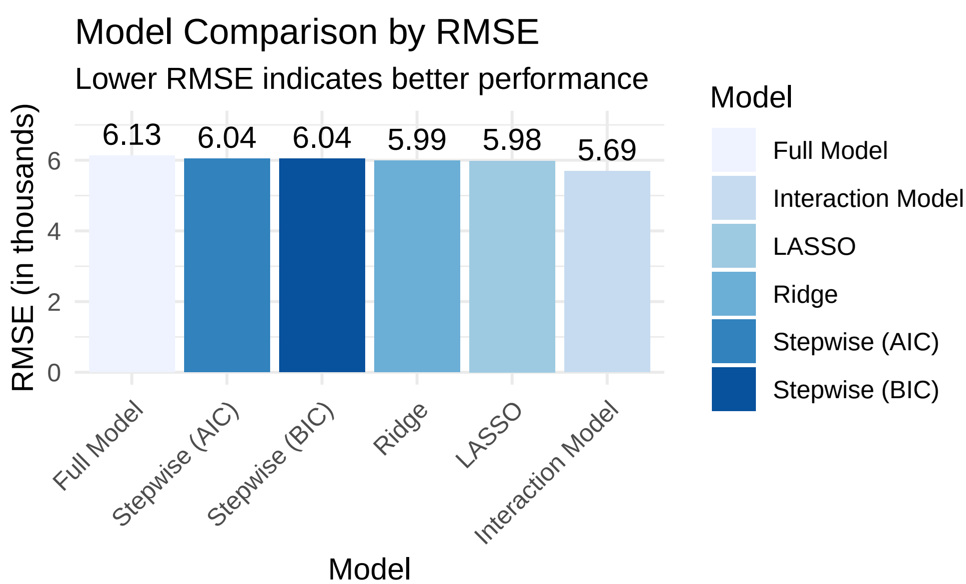

Figure 7: Model Comparison by RMSE

The interaction model performs best, reducing RMSE by $439.43.

6 Residual Diagnostics

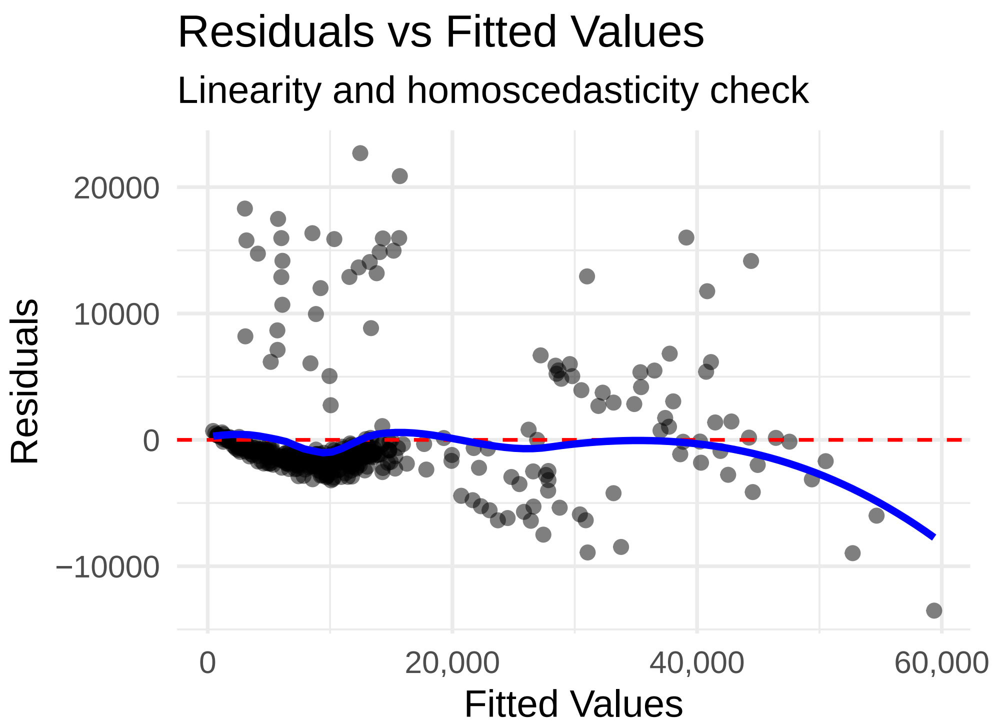

Figure 8: Residuals vs. Fitted Values

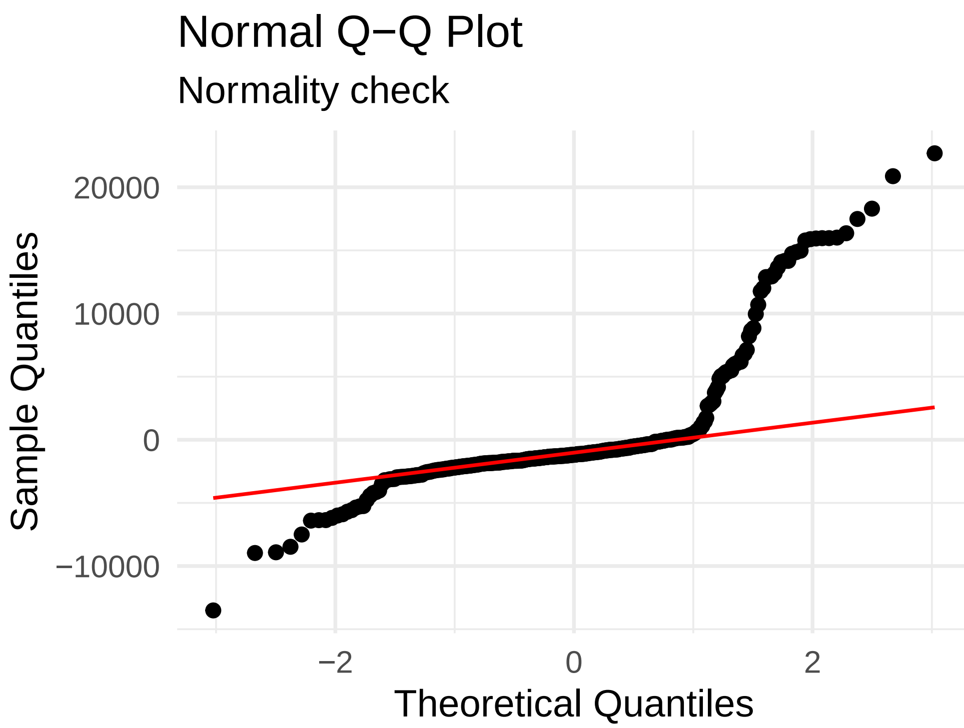

Figure 9: Normal Q-Q Plot

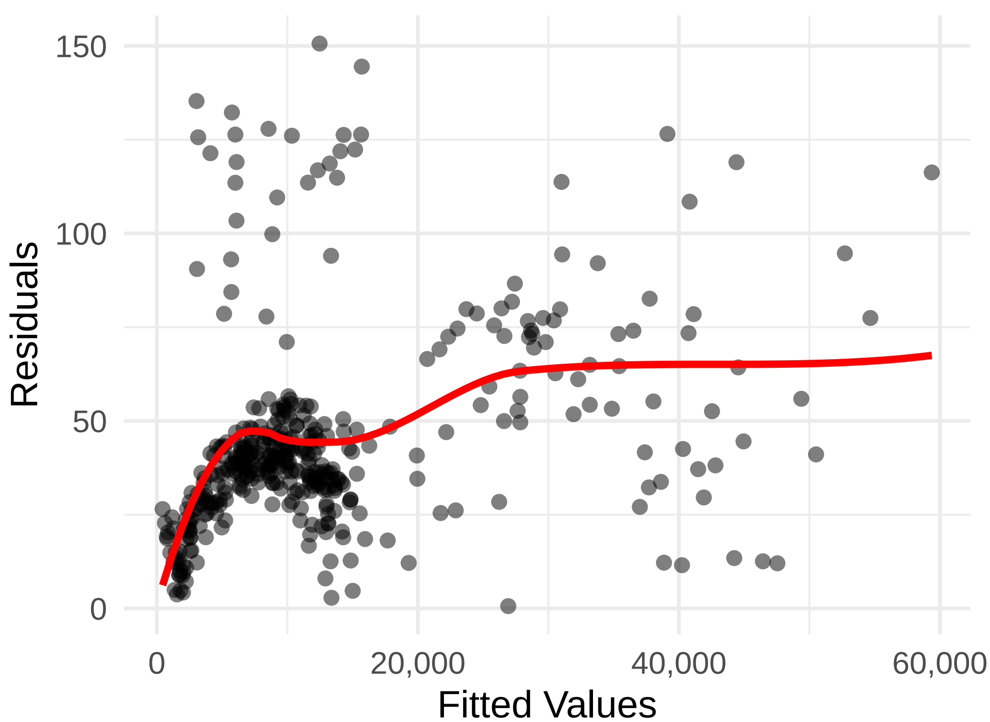

Figure 10: Scale-Location Plot

Diagnostics show acceptable linearity and normality, with some heteroscedasticity at higher values, suggesting potential log transformation or robust methods such as robust standard errors, weighted least squares, quantile regression, or generalised least squares (GLS) can be used to address heteroscedasticity when model residuals show increasing variance..

7 Conclusion

7.1 Key Findings

- **Best Model**: Interaction model, RMSE \$5,689, R^2 0.728
- **Key Predictors**: Smoking (\$20,000 increase), BMI-smoking interaction (6× stronger for smokers), age, children
- **Cost Implications**: A 50-year-old smoker (BMI 30) pays \$33,491 vs. \$9,952 as a non-smoker

7.2 Practical Implications

1. **Individuals**:
   - Quitting smoking offers the largest potential insurance savings
   - Weight management is particularly crucial for smokers, as the financial impact of BMI is much stronger
   - The combined effect of not smoking and maintaining healthy BMI could reduce annual costs by more than \$25,000 for some individuals
   - Age-related cost increases are inevitable but relatively modest compared to lifestyle factors
2. **Insurance Providers**:
   - Risk assessment should incorporate the interaction between smoking and BMI
   - More personalized premium structures could better reflect the compounding effects of multiple risk factors
   - Prevention programs focused on smoking cessation for overweight individuals could yield substantial cost savings
   - Predictive models should include interaction terms to accurately estimate costs
3. **Healthcare Policy**:
   - Public health initiatives targeting smoking cessation among overweight and obese individuals could yield disproportionately large healthcare savings
   - Programs might benefit from focusing efforts on regions with higher baseline costs
   - Weight management programs specifically designed for smokers could deliver enhanced value

7.3 Limitations and Future Work

1. **Data Limitations**:
   - The dataset may not capture all relevant factors affecting insurance costs
   - Additional variables like pre-existing conditions, medication usage, or healthcare utilization patterns would likely improve predictions
   - Cross-sectional data limits our ability to analyze how changes in risk factors affect charges over time
2. **Modeling Considerations**:
   - Log transformation of charges might better address heteroscedasticity
   - Additional interactions (e.g, age$\times$BMI) could potentially improve model performance
   - More sophisticated machine learning approaches might capture complex nonlinear relationships
   - Separate models for different regions or age groups might provide more tailored predictions
3. **Future Research Directions**:
   - Longitudinal studies to assess how changes in risk factors affect insurance costs over time
   - Analysis of cost-effectiveness of targeted interventions based on the interaction patterns identified
   - Exploration of additional interaction effects among demographic and health variables
   - Development of personalized risk calculators incorporating the identified factors

Our analysis provides valuable insights into the key drivers of health insurance costs, highlighting the dramatic impact of smoking, its interaction with BMI, and the significant influence of age. These findings can inform more effective risk assessment practices, precision-targeted public health interventions, and improved insurance pricing models, ultimately benefiting both individuals and healthcare systems.